Assignment 3 Workout

Thomas Vroom - i6291496

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Question 1

Let $X \sim \text{Beta}(6, 2)$. As calculated in the previous assignment, this means the PDF of X is equal to:

$$42x^5 - 42x^6$$

The CDF of X can be calculated as:

$$\int 42x^5 - 42x^6 \, dx = 7x^6 - 6x^7$$

Since this CDF has no gaps in it, we can calculate the median of X by setting the CDF equal to $\frac{1}{2}$:

$$7x^6 - 6x^7 = \frac{1}{2}$$

$$14x^6 - 12x^7 = 1$$

$$14x^6 - 12x^7 - 1 = 0$$

We can approximate the solution of this equation using numerical methods. If we do this we end up with the root:

$$x = 0.7715100364934$$

Thus we can conclude that $\text{Median}(X) = 0.7715100364934$. We can confirm this is true by looking at it graphically:

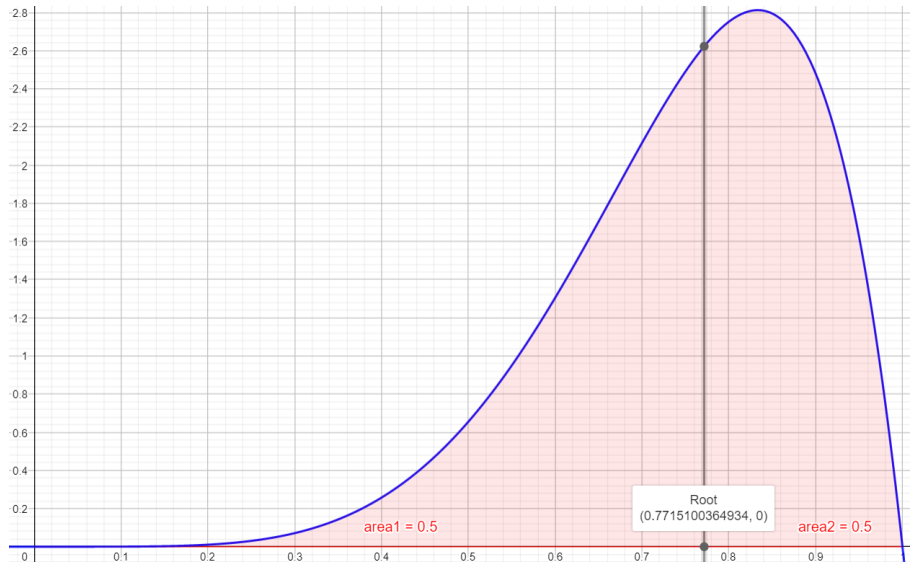

Figure 1: The PDF of X. As you can see the median divides the PDF into two perfect halves with an area of $\frac{1}{2}$

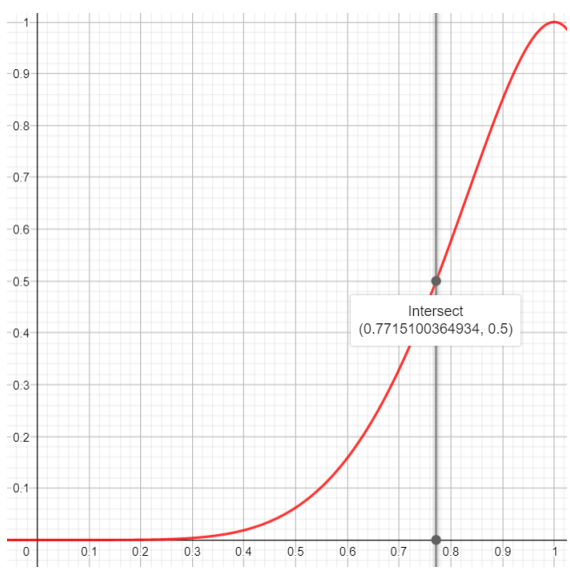

Figure 2: The CDF of X. As you can see the value of the CDF equals exactly $\frac{1}{2}$ at the median

Question 2

The histogram produced by the bean machine is created by dropping balls down multiple layers of pins laid out in a triangle shape. The probability of a ball going down the left or right side of a pin is $\frac{1}{2} = 0.5$, this makes this a **Bernoulli trial**. Because each ball falls down multiple layers of pins, each with a Bernoulli trial, the total fall from top to bottom is a string of repeated Bernoulli trials. This means that the path of each ball from top to bottom has a **binomial distribution** (repeated Bernoulli trials). The **Central Limit Theorem** states that given a large enough sample size (in this case, number of rows), the distribution approaches a normal distribution. This explains why the histogram created by dropping balls down many many rows of pins appears to have a normal distribution. In formulas:

Let X_i be a random variable that represents the final position of the ball. To get to this final position, X_i has to go through Bernoulli trials $Y_1, ..., Y_n$, where n represents the total number of rows in the bean machine.

$$X_i = Y_1 + ... + Y_n$$

$$\bar{X}_i = \frac{X_i}{n}$$

The central limit theorem states that:

$$\sqrt{n}\left(\frac{\bar{X}_n - \mu}{\sigma}\right) \sim \mathrm{N}(0,1)$$

From this we can deduce:

$$\sqrt{n}\left(\frac{\frac{X_i}{n} - \mu}{\sigma}\right) \sim \mathrm{N}(0,1)$$

$$\frac{\frac{X_i}{n} - \mu}{\sigma} \sim \frac{1}{\sqrt{n}} \cdot \mathrm{N}(0,1)$$

$$\frac{X_i}{n} - \mu \sim \frac{\sigma}{\sqrt{n}} \cdot \mathrm{N}(0,1)$$

$$\frac{X_i}{n} \sim \frac{\sigma}{\sqrt{n}} \cdot \mathrm{N}(0,1) + \mu$$

$$X_i \sim \frac{n \cdot \sigma}{\sqrt{n}} \cdot \mathrm{N}(0,1) + n \cdot \mu$$

$$X_i \sim \sigma\sqrt{n} \cdot \mathrm{N}(0,1) + n \cdot \mu$$

Which, by definition of the normal distribution, equals:

$$X_i \sim \mathrm{N}(n \cdot \mu, n \cdot \sigma^2)$$